Year 11 Set 2 · Higher · grade 5–6 · Lesson 3

Compound interest, depreciation & reverse percentages

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Today I will learn to...

Goals

1. Calculate compound interest and depreciation

2. Solve reverse and repeated percentage problems

Do now: retrieval practice

4 mins

1 Increase £200 by 5%

£210

2 Write a 15% decrease as a multiplier

0.85

3 Work out 1.1^2

1.21

4 After 20% off a price is £40. Original?

£50

Objective 1

Calculate compound interest and depreciation

Key idea

Learn

!

Find the multiplier, then use amount = original × multiplier^n, where n is the number of years. 4% interest → 1.04. 15% depreciation → 0.85.

Objective 1: Calculate compound interest and depreciation

I do: example 1

I do

£2 000 is invested at 3% compound interest for 4 years. Find the final amount.

1. Multiplier = 1.03
2. $2\,000 \times 1.03^4$
3. = £2 251.02

I do: example 2

I do

A car worth £15 000 depreciates by 12% a year. Find its value after 3 years.

1. Multiplier = 0.88
2. $15\,000 \times 0.88^3$
3. = £10 222.08

I do: example 3

I do

£500 earns 2% compound interest. How much interest is earned in 5 years?

1. Multiplier for 2% interest = 1.02
2. After 5 years: 500×1.02^5
3. = £552.04 (2 dp)
4. Interest = 552.04 − 500 = £52.04

Mini whiteboards

We do

1. £1 000 at 10% compound interest for 2 years

Answer: £1 210

2. Multiplier for 6% depreciation

Answer: 0.94

3. £500 at 2% compound for 3 years

Answer: £530.60

4. £800 depreciates 10% a year for 2 years

Answer: £648

You do: choose your challenge

You do

Bronze

£300 at 10% compound interest for 2 years

Silver

£2 500 at 4% compound interest for 3 years

Gold

A van worth £22 000 depreciates 18% a year. Find its value after 4 years (nearest £).

Platinum

Bank A: 3% for 5 years. Bank B: 3.5% in year 1, then 2.5% for 4 years. Which gives more on £10 000?

You do: answers

Check

Bronze	£363
Silver	£2 812.16
Gold	£9 947
Platinum	A: £11 592.74. B: £11 424.46. Bank A

If you finish early

Extra

Bronze

£200 at 5% compound interest for 2 years

£220.50

Silver

£4 000 at 3% compound interest for 4 years

£4 502.04

Gold

A car worth £15 000 depreciates 12% a year. Find its value after 3 years (nearest £).

£10 222

Platinum

£1 000 is invested at 4% compound interest. After how many whole years is it first worth over £1 200?

$1.04^4 \rightarrow$ £1 169.86, $1.04^5 \rightarrow$ £1 216.65. 5 years

Exam question 1

3 marks

Sam invests £4 000 at 2.5% compound interest per year. Work out the value after 3 years.

Space for working

Mark scheme: $4\,000 \times 1.025^3 = £4\,307.56$

Exam question 2

2 marks

A machine is worth £8 500. It loses 20% of its value each year. Find its value after 2 years.

Space for working

Mark scheme: $8\,500 \times 0.8^2 = £5\,440$

Objective 1 achieved

Calculate compound interest and depreciation



Objective 2

Solve reverse and repeated percentage problems

✓ Achieved: Calculate compound interest and depreciation

Key idea

Learn

!

For a reverse percentage, DIVIDE by the multiplier. For 'after how many years', keep multiplying until you pass the target.

Objective 2: Solve reverse and repeated percentage problems

I do: example 1

I do

After a 15% increase a price is £69. Find the original price.

1. Original = 100%; after a 15% increase = 115%
2. Multiplier = 1.15
3. Original × 1.15 = £69, so 69 ÷ 1.15
4. Original price = £60

I do: example 2

I do

After 2 years of 10% depreciation a bike is worth £405. Find its original value.

1. 10% depreciation → multiplier 0.9
2. 2 years → $0.9 \times 0.9 = 0.81$
3. Original × 0.81 = £405, so $405 \div 0.81$
4. Original value = £500

I do: example 3

I do

£1 000 earns 8% compound interest. After how many years is it first worth over £1 300?

1. $1\,000 \times 1.08^3 = £1\,259.71$
2. $1\,000 \times 1.08^4 = £1\,360.49$
3. 4 years

Mini whiteboards

We do

1. After 30% off, £56. Original price?

Answer: £80

2. After 5% interest, £420. Original amount?

Answer: £400

3. After 25% off, £75. Original price?

Answer: £100

4. After 3% interest, £515. Original amount?

Answer: £500

You do: choose your challenge

You do

Bronze

After 20% off, £36. Original price?

Silver

After a 12% increase, £84. Original price?

Gold

After 2 years of 5% compound interest a balance is £2 205. Find the starting amount.

Platinum

500 bacteria grow by 30% each hour. After how many whole hours will there first be more than 2 000?

You do: answers

Check

Bronze

£45

Silver

£75

Gold

$2\,205 \div 1.05^2 = £2\,000$

Platinum

$1.3^5 \rightarrow 1\,857$, $1.3^6 \rightarrow 2\,413$. 6 hours

If you finish early

Extra

Bronze

After 40% off, a coat costs £18. Find the original price.

£30

Silver

After a 15% increase, a price is £69. Find the original price.

£60

Gold

After 2 years of 10% compound growth a value is £605. Find the starting value.

605 ÷ 1.21 = £500

Platinum

A population of 8 000 falls by 6% a year. After how many whole years is it first below 6 000?

$0.94^4 \rightarrow 6\,246$, $0.94^5 \rightarrow 5\,871$. 5 years

Exam question 1

3 marks

A house rose in value by 6% to £318 000. Find its value before the rise.

Space for working

Mark scheme: 318 000 ÷ 1.06 = £300 000

Exam question 2

3 marks

A population of 2 000 grows by 5% each year. After how many years will it first exceed 2 400?

Space for working

Mark scheme: $2\,000 \times 1.05^3 = 2\,315$; $2\,000 \times 1.05^4 = 2\,431$; 4 years

Objective 2 achieved

Solve reverse and repeated percentage problems



Real life

Apply

A family buys a car for £18 000. It loses 15% of its value each year. Is it worth more than £10 000 after 3 years?

$18\,000 \times 0.85^3 = £11\,054.25$, so yes

Extension: stretch yourself

Stretch

1. An investment grows from £5 000 to £5 618 in 2 years at compound interest. Find the yearly rate.

$5\,618 \div 5\,000 = 1.1236 = 1.06^2$, so 6%

2. Which earns more: 4% compound interest for 3 years, or 12.5% simple interest over 3 years?

$1.04^3 = 1.1249 \rightarrow 12.49\%$, so the 12.5% simple interest earns slightly more

Exit ticket: exam questions

Exit

1. £3 000 is invested at 4% compound interest. Find its value after 2 years. **(2 marks)**

2. A phone cost £800 and loses 25% of its value each year. Find its value after 3 years. **(2 marks)**

3. After a 15% pay rise, Jo earns £27 600. What did she earn before? **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 $3\,000 \times 1.04^2 = £3\,244.80$ [2]

Q2 $800 \times 0.75^3 = £337.50$ [2]

Q3 $27\,600 \div 1.15 = £24\,000$ [3]

Total: _____ / 7

Today I achieved

- ✔ **Calculate compound interest and depreciation**
- ✔ **Solve reverse and repeated percentage problems**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. £1 500 is invested at 5% compound interest for 3 years. Find the final amount. [3]

2. A boat worth £40 000 depreciates by 10% a year. Find its value after 4 years. [2]

3. In a sale prices are cut by 15%. A coat costs £102 in the sale. Find the normal price. [3]

4. After a 3% increase, rent is £721 a month. Find the rent before the increase. [3]

5. £2 000 is invested at 7% compound interest. After how many whole years will it first be worth more than £2 600? [3]

Extra practice: Maths Genie → Edexcel question bank → Reverse Percentages · Compound Interest